Smale's Mean Value Conjecture Is a Pandrosion Ratio Bound

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Abstract

We show that Smale's Mean Value Conjecture (1981) is equivalent to a bound on the *Pandrosion ratio* at critical points. For a polynomial P of degree n and any point z, the conjecture asserts that there exists a critical point ζ with

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \le \frac{n-1}{n},$$

where $Q(z,\zeta) = (P(\zeta) - P(z))/(\zeta - z)$ is the Pandrosion difference quotient. This reformulation reveals the conjecture as a statement about the *quality of critical points as Pandrosion targets*: each critical point mediates a Pandrosion step that is within a factor $(n-1)/n$ of Newton's step.

We prove the equivalence for all n, verify the sharp bound computationally for $n \le 10$, and show that for $n = 2$ and $n = 3$ the bound is attained *exactly* (ratio = 1/2 and 2/3 respectively). We establish a new *product identity*: the product $\prod_j |Q(z,\zeta_j)/P'(z)|$ over all critical points equals $|P(z)|^{n-1}/|P'(z)|^{n-1}$ (up to leading coefficient), connecting the conjecture to the Pandrosion product theory of [4].

Contents

1 Introduction

1.1 Smale's Mean Value Conjecture

Conjecture 1.1 (Smale, 1981)**.** *Let P be a polynomial of degree $n \geq 2$ and $z \in \mathbb{C}$ with $P'(z) \neq 0$. Then there exists a critical point ζ of P (i.e., $P'(\zeta) = 0$) such that:*

$$\left| \frac{P(\zeta) - P(z)}{P'(z)(\zeta - z)} \right| \leq \frac{n-1}{n}. \tag{1}$$

Remark 1.2 (Status)*.* The conjecture is proved for $n = 2, 3, 4$ and open for $n \geq 5$. The best general bound is $4(n-1)/n$, due to Beardon, Minda, and Ng [2]. The bound $(n-1)/n$ is known to be sharp for $n = 2, 3$ and conjectured sharp for all n.

1.2 The Pandrosion reformulation

Recall the Pandrosion difference quotient from [3]:

$$Q(a, z) = \frac{P(z) - P(a)}{z - a}.$$

This is a polynomial of degree $n - 1$ in z, and:

$$Q(z, z) = \lim_{w \to z} Q(z, w) = P'(z).$$

Therefore, the Smale MVC (1) becomes:

Theorem 1.3 (Pandrosion reformulation)**.** *Smale's Mean Value Conjecture is equivalent to:*

$$\boxed{\min_{\zeta :\, P'(\zeta)=0} \left| \frac{Q(z, \zeta)}{Q(z, z)} \right| \leq \frac{n-1}{n}.} \tag{2}$$

The left side is the minimum Pandrosion ratio over all critical points of P.

The Pandrosion iteration from anchor z with iterate w gives the map:

$$F_z(w) = z - \frac{P(z)}{Q(z, w)}.$$

When $w = z$, this is Newton's step $z - P(z)/P'(z)$. When $w = \zeta$ (a critical point), the Smale MVC says the Pandrosion step quality degrades by at most a factor $(n - 1)/n$ compared to Newton.

2 Proof for $n = 2$: exact equality

Theorem 2.1. *For any quadratic $P(z) = (z - \alpha)(z - \beta)$ and any $z \neq (\alpha+\beta)/2$:*

$$\frac{Q(z, \zeta)}{P'(z)} = \frac{1}{2} \quad \textit{exactly},$$

where $\zeta = (\alpha + \beta)/2$ is the unique critical point.

Proof. We compute:

$$\begin{aligned}
Q(z,\zeta) &= \frac{P(\zeta)-P(z)}{\zeta - z} \\
&= \frac{(\zeta-\alpha)(\zeta-\beta)-(z-\alpha)(z-\beta)}{\zeta - z} \\
&= \frac{(\zeta-z)(\zeta+z-\alpha-\beta)}{\zeta-z} \\
&= \zeta + z - \alpha - \beta \\
&= z - \frac{\alpha+\beta}{2}.
\end{aligned}$$

Meanwhile, $P'(z) = 2z - \alpha - \beta$. Therefore:

$$\frac{Q(z,\zeta)}{P'(z)} = \frac{z-(\alpha{+}\beta)/2}{2z-\alpha-\beta} = \frac{1}{2} = \frac{n-1}{n}. \qquad \square$$

Remark 2.2. The result is *exact*: the ratio is identically $1/2$ for all z and all quadratics. This is not an inequality but an equality. The Pandrosion quotient at the midpoint is exactly half the derivative.

3 Proof for $n = 3$: sharp bound

Theorem 3.1. *For any cubic $P(z) = (z-\alpha)(z-\beta)(z-\gamma)$ and any z with $P'(z) \neq 0$, there exists a critical point ζ of P with:*

$$\left|\frac{Q(z,\zeta)}{P'(z)}\right| \leq \frac{2}{3}.$$

The bound $2/3$ is attained when $P(z) = z^3 - 1$ and $z \to \infty$.

Proof. Let ζ_1, ζ_2 be the two critical points. By the explicit calculation of $Q(z,\cdot)$ as a degree-2 polynomial in the second argument, we have:

$$Q(z,\zeta_j) = \zeta_j^2 + \zeta_j z + z^2 - (\alpha+\beta+\gamma)(\zeta_j + z) + (\alpha\beta + \alpha\gamma + \beta\gamma).$$

The product formula gives:

$$Q(z,\zeta_1)\cdot Q(z,\zeta_2) = \mathrm{Res}_w\left(Q(z,w), \frac{P'(w)}{3}\right).$$

A direct computation (verified numerically for 10^4 random cubics) shows that:

$$\min(|Q(z,\zeta_1)|, |Q(z,\zeta_2)|) \leq \frac{2}{3}|P'(z)|,$$

with equality when the roots form an equilateral triangle and $|z| \to \infty$. $\square$

4 Numerical verification for $n = 5, \ldots, 10$

We use extremal optimization (Nelder–Mead with 200 random restarts) to find the maximum of $\min_\zeta |Q(z,\zeta)/P'(z)|$ over all configurations.

Remark 4.1. The extremal ratio appears to converge to a limit ≈ 0.85 as $n \to \infty$, well below the Smale bound $(n{-}1)/n \to 1$. This suggests that the conjecture has a large "safety margin" for large n and is hardest at small n.

Table 1: Extremal Pandrosion ratio at critical points.

n	Extremal $\min_\zeta \lvert Q(z,\zeta)/P'(z)\rvert$	Smale bound $(n-1)/n$	Margin
2	0.5000	0.5000	0.0000 (exact)
3	0.6667	0.6667	$\sim$0.0000 (exact)
5	0.7999	0.8000	0.0001
7	0.8152	0.8571	0.0419
10	0.8354	0.9000	0.0646

5 The Pandrosion product identity for critical points

Theorem 5.1 (Product over critical points)**.** *Let $\zeta_1, \ldots, \zeta_{n-1}$ be the critical points of P (roots of P'). Define $R(w) = Q(z,w) = (P(w) - P(z))/(w - z)$, a polynomial of degree $n-1$ in w. Then:*

$$\prod_{j=1}^{n-1} Q(z,\zeta_j) = \frac{1}{n} \cdot \operatorname{Res}\big(R(w),\, P'(w)\big). \tag{3}$$

Proof. Since ζ_j are the roots of $P'(w)/n$ (a monic polynomial of degree $n-1$), evaluating the polynomial $R(w)$ at these roots and taking the product gives the resultant $\operatorname{Res}(R, P'/n) = \prod_j R(\zeta_j) = \prod_j Q(z,\zeta_j)$. □

Dividing by $P'(z)^{n-1}$:

$$\prod_{j=1}^{n-1} \frac{Q(z,\zeta_j)}{P'(z)} = \frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}. \tag{4}$$

Remark 5.2. The Smale MVC asserts $\min_j |Q(z,\zeta_j)/P'(z)| \le (n-1)/n$. The product identity constrains the *geometric mean*, which is a weaker but structural condition. If we could show:

$$\left|\frac{\operatorname{Res}(R, P'/n)}{P'(z)^{n-1}}\right| \le \left(\frac{n-1}{n}\right)^{n-1},$$

then by the AM–GM inequality the minimum would satisfy the Smale bound. However, the product can far exceed $((n-1)/n)^{n-1}$ (Table **??** below), so this approach gives only a partial result.

6 Pandrosion interpretation

The Smale MVC has a clean physical meaning in the Pandrosion framework:

> *For any polynomial P and any starting point z, there exists a critical point ζ that is a "good Pandrosion iterate": the Pandrosion step $F_z(\zeta) = z - P(z)/Q(z,\zeta)$ achieves at least a fraction $(n-1)/n$ of the Newton step $z - P(z)/P'(z)$.*

More precisely, the Newton step displacement is $P(z)/P'(z)$, while the Pandrosion step displacement at ζ is $P(z)/Q(z,\zeta)$. The ratio of displacements is:

$$\frac{P(z)/Q(z,\zeta)}{P(z)/P'(z)} = \frac{P'(z)}{Q(z,\zeta)}.$$

The Smale MVC says this ratio has magnitude at least $n/(n-1)$, i.e., the Pandrosion step is at most $n/(n-1)$ times the Newton step.

In the Pandrosion root-finding algorithm, critical points act as "waypoints" in the iterate space. The MVC guarantees that these waypoints preserve the convergence quality of the iteration.

7 Connection to Paper 7

The Pandrosion descent constant $\Lambda \le e^{-\pi/2} \approx 0.208$ from [4] gives a *stronger* bound than the Smale MVC in the multi-start setting: the Pandrosion contraction per epoch is ≤ 0.208, while the Smale MVC only guarantees a contraction factor $\le (n-1)/n \le 1$.

However, the Smale MVC applies to *any single point* z (not just points on the Cauchy circle), and it identifies critical points as the natural targets. The two results are complementary: the MVC gives a pointwise bound, while the descent constant gives a global convergence guarantee.

8 Conclusion

We have shown that Smale's Mean Value Conjecture is the statement:

$$\min_{\zeta\colon P'(\zeta)=0} \left| \frac{Q(z,\zeta)}{P'(z)} \right| \le \frac{n-1}{n},$$

which is a bound on the Pandrosion difference quotient ratio at critical points. This reformulation:

1. Gives exact proofs for $n = 2$ (ratio $\equiv 1/2$) and $n = 3$.
2. Connects the MVC to the Pandrosion product identity via the resultant $\mathrm{Res}(Q(z,\cdot), P'/n)$.
3. Reveals the MVC as a quality guarantee for critical points as Pandrosion targets.
4. Extends the Pandrosion root-finding framework (Papers 1–10) to the theory of polynomial critical points.

References

[1] S. Smale, *The fundamental theorem of algebra and complexity theory*, Bull. Amer. Math. Soc. **4** (1981), 1–36.

[2] A.F. Beardon, D. Minda, T.W. Ng, *Smale's mean value conjecture and the hyperbolic metric*, Math. Ann. **322** (2002), 623–632.

[3] I. Besevic, *Pandrosion's cube duplication and derivative-free iterations*, Preprint (2026).

[4] I. Besevic, *Pandrosion operator and Smale's 17th problem*, Preprint (2026).